Experiment 3

Priority Encoder and Push-Button Polarity

Student	Seyed Kiarash Eshraghi Naeini (403101383)
Instructor	Dr. Mohammad Reza Movahhedin Teaching assistant: Hossein Moghim
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Platform	Zynq-7010 (XC7Z010-CLG400-1), 50 MHz clock, Vivado 2019.1

Summary

A 4-input priority encoder is implemented on the FPGA with the board push buttons as inputs and the board LEDs as outputs. The first version treats the buttons as active-high; the second inverts them, because the buttons of the board read 0 when pressed. Both were verified exhaustively for this report. Code: `https://github.com/seyedkiarash/seyedkiarash.github.io/tree/main/labs/logic-circuits-lab/lab03-priority-encoder`.

1 Design

A priority encoder outputs the index of the highest-priority active input. Here the output is the button number *plus one*, so that 000 can mean "no button pressed": The priority is expressed directly by an `if...else`

button[3]	button[2]	button[1]	button[0]	LED[2:0]
1	×	×	×	100
0	1	×	×	011
0	0	1	×	010
0	0	0	1	001
0	0	0	0	000

`if` chain, which synthesis turns into the equations $L_2 = b_3$, $L_1 = \overline{b_3}(b_2 + b_1)$, $L_0 = \overline{b_3}(b_2 + \overline{b_1}b_0)$ (2 LUTs).

```
always @(*) begin
    if      (A[3]) c = 3'b100;
    else if (A[2]) c = 3'b011;
    else if (A[1]) c = 3'b010;
    else if (A[0]) c = 3'b001;
    else           c = 3'b000;
end
```

2 Part 3 vs. Part 5 – button polarity

In Part 3 the buttons were connected directly. On the board, however, a button reads *0* when it is pressed, so with no button pressed all inputs are 1 and the encoder shows 100. Part 5 places the same encoder (`encoderpr`) behind an inverter, `.A(~button)`, after which the LEDs behave as intended. The same inversion is used in the later experiments wherever the board buttons serve as data inputs.

3 Verification

Testbench	What is checked	Result
`tb_priority_encoder`	all 16 input combinations, active-high	**PASS**
`tb_part5`	all 16 combinations, active-low buttons	**PASS**

4 Conclusion

The priority encoder is a direct translation of its truth table into a prioritised conditional chain. The second version is a reminder to check the electrical polarity of every input before testing logic on the board.